

Supplementary Material for Advancing Stress Detection with Chaotic Attractor Informed Synthesis of PPG Signal

APPENDIX A IMPLEMENTATION DETAILS

The PPG signals undergo an initial interpolation process in our experiments to achieve a uniform sampling rate of 32 Hz. Subsequently, a third-order Butterworth filter is used to eliminate high-frequency and low-frequency noise, where the filtering frequency is set to 0.5-8 Hz. Finally, each signal is divided into time slices of 6 seconds with an overlap of 2 seconds using a sliding window. The reason for selecting the specific length of the sample is based on the fact that when analyzing PPG using PSR and Recurrence Plot, we found that signals shorter than 5s cannot reach the desired density of the attractor. To balance the good desired density of attractor against the need for near real-time pressure detection, we choose 6s as the sample length.

The proposed model comprises two modules: the attractor extraction network and the generative adversarial network. The central component of the attractor extraction network is constructed using a single layer of LSTM units, and the technique of TF is employed to impose constraints on the model. The maximum Lyapunov exponent of the PPG signal is computed to be 0.029, leading to the determination of the constrained interval as $\tau = 12$. In the NODE implementation for updates between hidden states, we utilize a fully connected network to parameterize ordinary differential equations and choose the Runge-Kutta fourth-order method as the numerical solver. In the generative adversarial network, the generator and discriminator consist of fully connected layers with the Leaky ReLU activation function. The dimension of the input noise vector in the generator is equal to the sampling rate of signals. We set $\lambda = 1$ and $\eta = 10$ where η is the default setting of WGAN-GP. The model is implemented within the PyTorch framework, and all the components are optimized using the Adam optimizer with a learning rate of 0.001. Experimental results are obtained on a workstation with an Intel Xeon E7-4820 v4 CPU, 256 GB of RAM, and an NVIDIA Tesla M60 GPU.

APPENDIX B MODEL FRAMEWORK DETAILS

TABLE I: Details of the attractor extraction network.

Input	Operation	Output
$x_{all} : (BS, T, 1)$		
$x_0 : (BS, 1, 1)$	<i>Encoder : Linear</i> ($BS, 1, z$)	$z_0 : (BS, 1, z)$
$z_0 : (BS, 1, z)$	<i>Node</i> ($z, 2 * z, n_{step}$), <i>Linear</i> ($2 * z, z$)	$z'_0 : (BS, 1, z)$
$z'_0 : (BS, 1, z)$	<i>LSTM</i> ($BS, 1, z$)	$z_1 : (BS, 1, z)$
$z_1 : (BS, 1, z)$	<i>Node</i> ($z, 2 * z, n_{step}$), <i>Linear</i> ($2 * z, z$)	$z'_1 : (BS, 1, z)$
$z'_1 : (BS, 1, z)$	<i>LSTM</i> ($BS, 1, z$)	$z_2 : (BS, 1, z)$
$x_2 : (BS, 1, 1)$	<i>Encoder : Linear</i> ($1, z$)	$z_2 : (BS, 1, z)$
$z_2 : (BS, 1, z)$	<i>Node</i> ($z, 2 * z, n_{step}$), <i>Linear</i> ($2 * z, z$)	$z'_2 : (BS, 1, z)$
$z'_2 : (BS, 1, z)$	<i>LSTM</i> ($BS, 1, z$)	$z_3 : (BS, 1, z)$
...	...	...
$z_{T-1} : (BS, 1, z)$	<i>Node</i> ($z, 2 * z, n_{step}$), <i>Linear</i> ($2 * z, z$)	$z'_{T-1} : (BS, 1, z)$
$z'_{T-1} : (BS, 1, z)$	<i>LSTM</i> ($BS, 1, z$)	$z_T : (BS, 1, z)$
$z_{all} : (BS, T, z)$	<i>Decoder : Linear</i> ($BS, z, 1$)	$x'_{all} : (BS, T, 1)$

TABLE II: Details of the discriminator network.

Input	Operation	Output
$(BS, 1, T)$	<i>Linear</i> ($T, 512$), <i>LeakyReLU</i> , <i>Dropout</i> (0.3)	$(BS, 1, 512)$
$(BS, 1, 512)$	<i>Linear</i> ($512, 1024$), <i>LeakyReLU</i> , <i>Dropout</i> (0.3)	$(BS, 1, 1024)$
$(BS, 1, 1024)$	<i>Linear</i> ($1024, 1280$), <i>LeakyReLU</i> , <i>Dropout</i> (0.3)	$(BS, 1, 1280)$
$(BS, 1, 1280)$	<i>Linear</i> ($1280, 1280$), <i>LeakyReLU</i> , <i>Dropout</i> (0.3)	$(BS, 1, 1280)$
$(BS, 1, 1280)$	<i>Linear</i> ($1280, 256$), <i>LeakyReLU</i> , <i>Dropout</i> (0.3)	$(BS, 1, 256)$
$(BS, 1, 256)$	<i>Linear</i> ($256, 1$)	$(BS, 1, 1)$

TABLE III: Details of the generator network.

Input	Operation	Output
$(BS, 1, 32)$	$Linear(32, 192), LeakyReLU, Dropout(0.3)$	$(BS, 1, 192)$
$(BS, 1, 192)$	$Linear(192, 256), LeakyReLU, Dropout(0.3)$	$(BS, 1, 256)$
$(BS, 1, 256)$	$Linear(256, 512), LeakyReLU, Dropout(0.3)$	$(BS, 1, 512)$
$(BS, 1, 512)$	$Linear(512, 1024), LeakyReLU, Dropout(0.3)$	$(BS, 1, 1024)$
$(BS, 1, 1024)$	$Linear(1024, 256), LeakyReLU, Dropout(0.3)$	$(BS, 1, 256)$
$(BS, 1, 256)$	$Linear(256, 256), LeakyReLU, Dropout(0.3)$	$(BS, 1, 256)$
$(BS, 1, 256)$	$Linear(256, T), Tanh$	$(BS, 1, T)$

In this section, the architecture details of the attractor extraction network are shown in table I. The attractor extraction network receives an input vector in the $(BS, 1, T)$ format, where BS and T are defined as above. The network is comprised of a single-layer *LSTM* unit with dimension z , an encoder-decoder structure that is a fully connected network, and a *NODE*. The *NODE* is parameterized by a fully connected network, and its time step is set to n_{step} . The network's gradient is truncated regularly using the Teacher Forcing (TF) technique, specifically every τ time unit. In this case, τ is set to 2 for illustrative purposes.

The architecture details of the discriminator are shown in table II. The discriminator receives an input vector in the format of $(BS, 1, T)$, where BS is the batch size, 1 is the number of channels, and the number of time points T is a 6-second time window multiplied by the sampling rate of the PPG signals. The networks consist of fully connected layers with the *LeakyReLU* activation function.

The architecture details of the generator are shown in table III. The generator receives an input vector in the format of $(BS, 1, 32)$, where BS is the batch size, 1 is the number of the channels and the number of time points, 32 is the dimension of the Gaussian noise, which is equal to the sampling rate of signals. The network consists of fully connected layers with the *LeakyReLU* activation function. The final output dimension T is a 6-second time window multiplied by the sampling rate of the PPG signals and uses tanh as the activation function.

APPENDIX C

LORENZ SYSTEM AND RÖSSLER SYSTEM

The Lorenz system is a system of ordinary differential equations first studied by Edward Lorenz. The chaotic attractor of the Lorenz system, known as the "Lorenz attractor," exhibits a distinctive "butterfly" or "figure-eight" pattern in phase space. This structure displays the irregular yet patterned transitions between states of the system. The Lorenz system is described by the following equations:

$$\begin{aligned}
 \frac{dx}{dt} &= \sigma(y - x), \\
 \frac{dy}{dt} &= x(\rho - z) - y, \\
 \frac{dz}{dt} &= xy - \beta z,
 \end{aligned} \tag{1}$$

where σ , ρ , and β are the system parameters. Typical values at which the Lorenz system shows a chaotic behavior are $\sigma = 10$, $\rho = 28$, and $\beta = \frac{8}{3}$.

The Rössler system is another system that exhibits chaotic dynamics. A notable feature of the Rössler system is that its chaotic attractor tends to have a simpler structure, often showing clear spiral patterns. The Rössler system is described by the following equations:

$$\begin{aligned}
 \frac{dx}{dt} &= -y - z, \\
 \frac{dy}{dt} &= x + ay, \\
 \frac{dz}{dt} &= b + z(x - c),
 \end{aligned} \tag{2}$$

where a , b , and c are the system parameters. Typical values at which the Rössler system shows a chaotic behavior are $a = 0.2$, $b = 0.2$, and $c = 5.7$.

APPENDIX D

ADDITIONAL EXPERIMENT RESULTS ON UBFC AND CLAS DATASETS

In this section, Fig. 1 illustrates the waveform of real and generated signals on UBFC and CLAS datasets. For each dataset, the first row corresponds to non stress condition and the second row corresponds to stress condition.

Fig. 2 shows the visualization of the feature distribution matching between real and generated signals on UBFC and CLAS datasets, where red indicates the distribution of real signals and blue indicates the distribution of generated signals. For each dataset, the first row corresponds to non stress condition and the second row corresponds to stress condition.

table IV contains the quantitative assessment results of similarity on UBFC and CLAS datasets. Our proposed method achieves the best results in most situations.

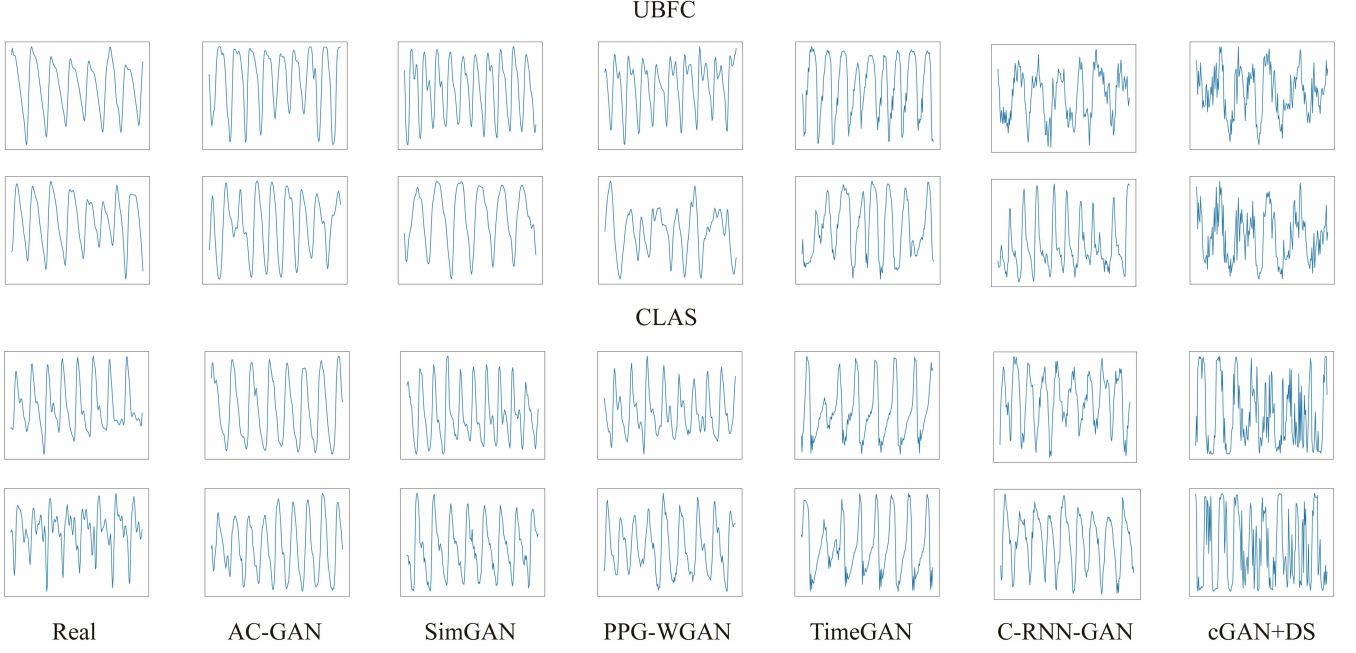


Fig. 1: Waveform of real and generated signals on the UBFC and CLAS datasets.

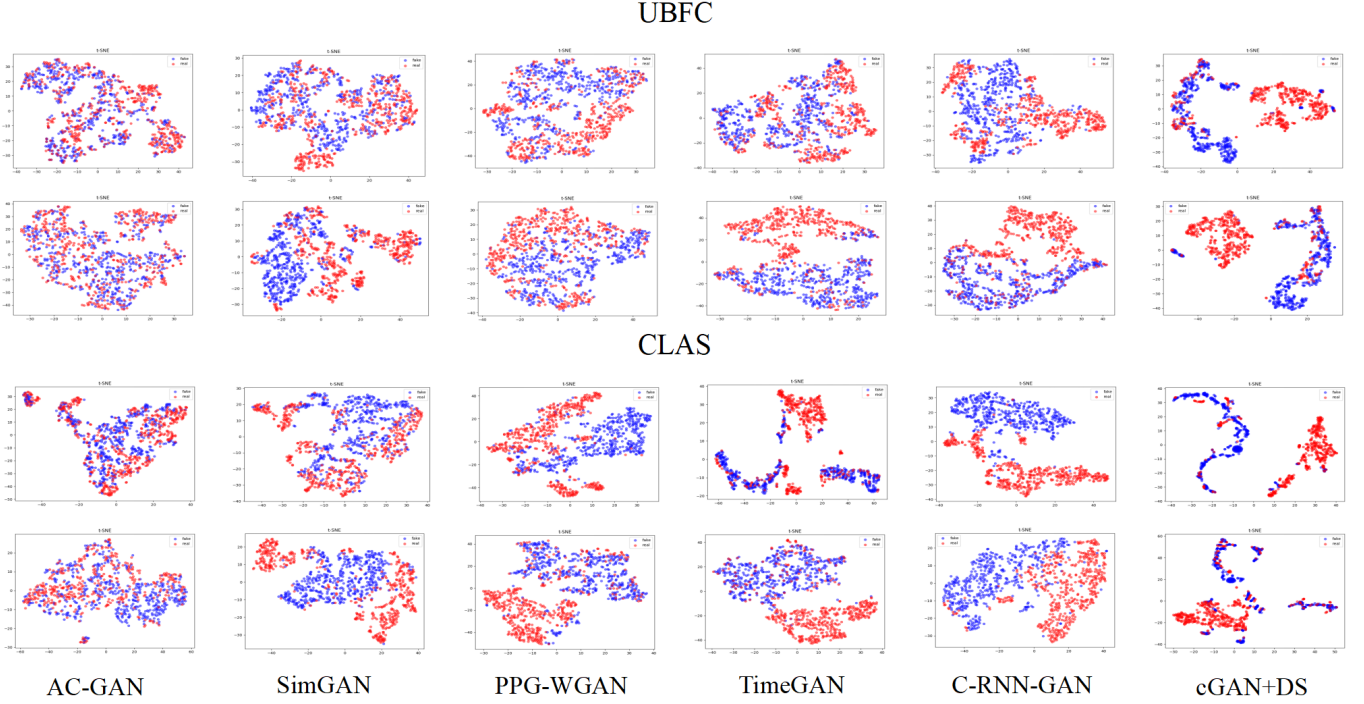


Fig. 2: Visualization of feature distribution on the UBFC and CLAS datasets.

TABLE IV: Quantitative assessment results of similarity on UBFC and CLAS datasets. The $\downarrow$ indicates “the smaller the better”, and $\uparrow$ indicates “the larger the better”. **Bold** indicates the best score in each row.

Dataset	State	Measure	AC-GAN	SimGAN	PPG-WGAN	TimeGAN	C-RNN-GAN	cGAN+DS
UBFC	Non-stress	MMD($\downarrow$)	0.2362	0.2466	0.3374	1.7292	1.0721	2.7357
		MSE($\downarrow$)	0.5754	0.5813	0.5784	0.5042	0.7951	0.5978
		DTW($\downarrow$)	6.92	6.97	6.69	6.50	8.16	7.03
		MIC($\uparrow$)	0.563	0.564	0.567	0.560	0.493	0.521
	Stress	MMD($\downarrow$)	0.2076	0.3392	1.181	2.2063	1.8243	2.7138
		MSE($\downarrow$)	0.4041	0.5215	2.0105	0.4691	0.6112	0.4782
		DTW($\downarrow$)	5.79	6.58	11.91	6.20	7.04	6.28
		MIC($\uparrow$)	0.564	0.562	0.554	0.559	0.516	0.537
CLAS	Non-stress	MMD($\downarrow$)	0.1862	0.3044	0.2699	2.4514	1.8823	2.4849
		MSE($\downarrow$)	0.5186	0.5296	0.5317	0.6713	0.6360	0.7186
		DTW($\downarrow$)	6.56	6.64	6.65	7.28	7.18	7.78
		MIC($\uparrow$)	0.577	0.583	0.575	0.817	0.505	0.556
	Stress	MMD($\downarrow$)	0.2770	0.2730	0.4063	2.0881	0.4430	2.0847
		MSE($\downarrow$)	0.5493	0.5436	0.6008	0.5751	0.5843	0.7129
		DTW($\downarrow$)	6.76	6.73	7.06	6.91	6.99	7.75
		MIC($\uparrow$)	0.586	0.584	0.598	0.580	0.520	0.535

APPENDIX E ADDITIONAL RESULT OF USABILITY AND SIMILARITY RANKING

To comprehensively analyse and evaluate the experimental results, we employed the Nemenyi post-hoc test method, with the results presented in Fig. 3.

Fig. 3a shows the rankings of different models in terms of usability, which aims to assess the contribution of the data generated by each model to the accuracy of the stress classification task. As can be clearly seen from the figure, our AC-GAN model ranks first, demonstrating its superiority in generating the most valuable data for downstream classification tasks.

Additionally, in terms of similarity rankings, as shown in Fig. 3b, we used distance metrics to assess the similarity between the signals generated by each model and the real signals. AC-GAN also achieved the best performance, with the signals it generated being the closest to the real PPG signals.

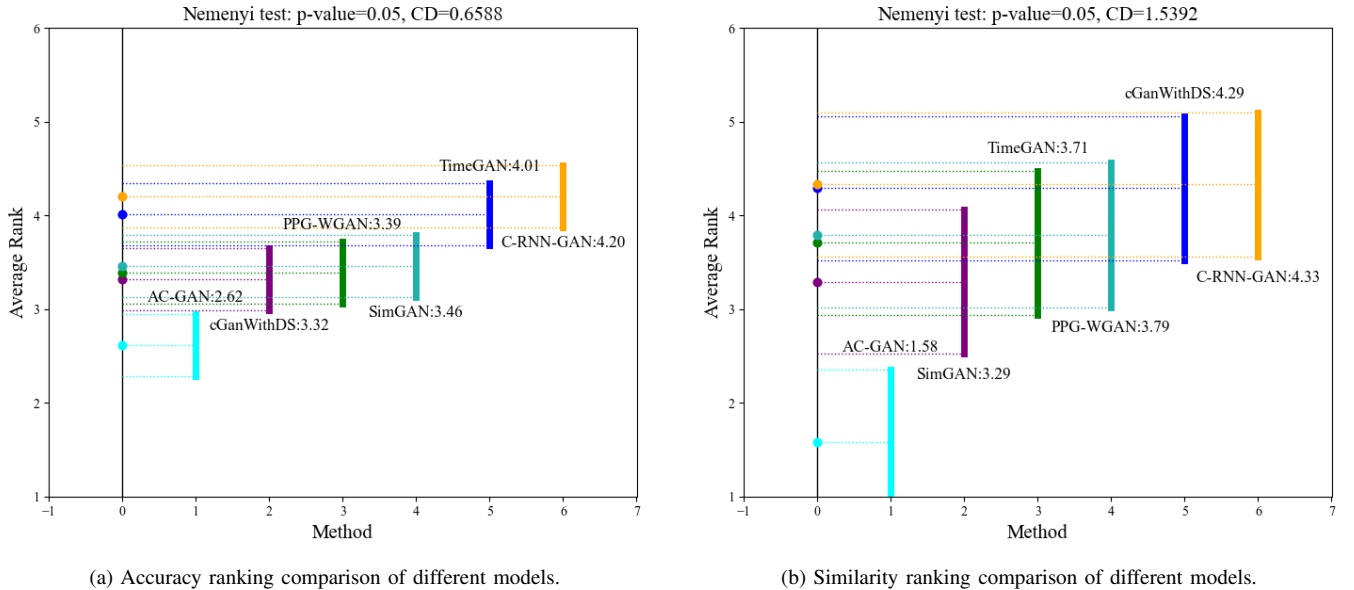


Fig. 3: Comparison of accuracy and similarity rankings across models.

APPENDIX F ADDITIONAL RESULT OF ABLATION STUDY

Fig. 4 shows the visualization of the feature distribution matching between real signals and signals generated by AC-GAN, WGAN-GP without Attractor Constraint and AC-GAN without NODE mechanism. The feature distribution is visualized using

t-SNE, where red indicates the distribution of real signals and blue indicates the distribution of generated signals.

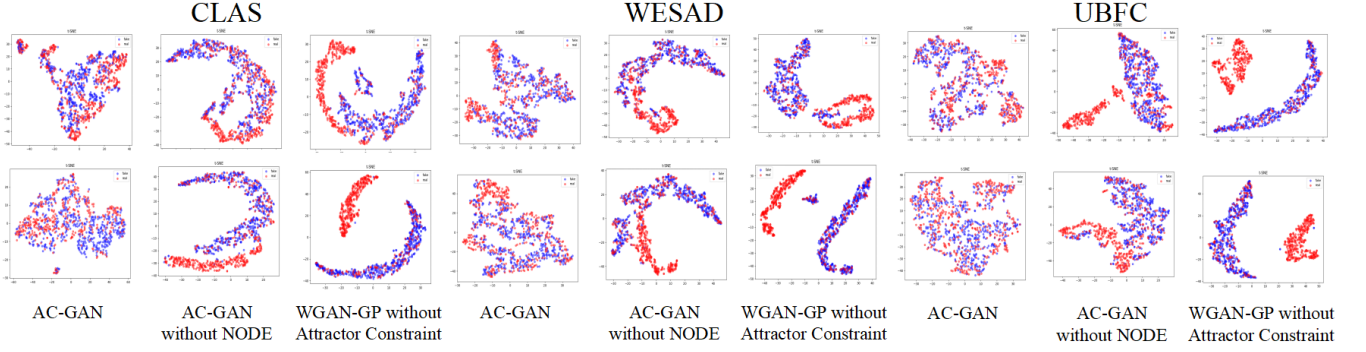


Fig. 4: Visualization of the feature distribution in ablation experiments. The first row corresponds to non stress condition and the second row corresponds to stress condition.

table V shows the quantitative assessment results of similarity on UBFC and CLAS datasets in the ablation study. The proposed AC-GAN model outperforms the ablated models in all cases.

TABLE V: Quantitative assessment results of similarity on UBFC and CLAS datasets in ablation study.

Dataset	State	Measure	AC-GAN	AC-GAN without NODE	WGAN-GP without Attractor Constraint
UBFC	Non-stress	MMD($\downarrow$)	0.2362	0.2430	0.2627
		MSE($\downarrow$)	0.5754	0.5780	0.5985
		DTW ($\downarrow$)	6.92	6.98	7.06
		MIC($\uparrow$)	0.563	0.560	0.554
	Stress	MMD($\downarrow$)	0.2076	0.216	0.2628
		MSE($\downarrow$)	0.4041	0.4147	0.4340
		DTW($\downarrow$)	5.79	5.87	6.02
		MIC($\uparrow$)	0.564	0.555	0.554
CLAS	Non-stress	MMD($\downarrow$)	0.1862	0.1892	0.2050
		MSE($\downarrow$)	0.5186	0.5318	0.5473
		DTW($\downarrow$)	6.56	6.65	7.74
		MIC($\uparrow$)	0.577	0.575	0.571
	Stress	MMD($\downarrow$)	0.2770	0.2786	0.2833
		MSE($\downarrow$)	0.5493	0.5539	0.6008
		DTW($\downarrow$)	6.76	6.81	6.81
		MIC($\uparrow$)	0.586	0.584	0.580